PPE Compliance Detection — Team Guide

Written so that every team member can understand the project and answer the judges, even without a coding background.
Project folder: `D:\software\ppe-detection` • 30 Oct 2026

> **How to use this guide.** Read sections 1–5 together (about 20 minutes). Then each person takes one **role** from section 9 and practises that role's questions out loud. If a judge asks you something outside your role, it is fine to say "*[teammate] handled that part, I'll let them answer*" — but you should still be able to explain sections 1–3 yourself.

1. The problem, and what we built

The problem (from the problem statement): supervisors cannot watch every worker in a busy industrial area, so workers without helmets or safety vests are missed.

What our system does: you give it a photo (or a short video). For **each worker** it decides: *is this worker wearing a helmet? a vest?* It then draws a coloured box around the worker and writes the verdict, lists what is missing, and creates a summary.

Colour on the image	Meaning	Example label
GREEN	Worker has everything required	Worker 1: OK
RED	Worker is missing something	Worker 2: NO HELMET + NO VEST
AMBER	Cannot be sure (too small, cut off, crowded, borderline) — a human should look	Worker 3: REVIEW

> **The most important idea in the whole project.** The problem statement says: *"A helmet detected anywhere in the image must not be taken to mean every worker is compliant."* So we do not just ask "is there a helmet in the picture?" We check **whose head the helmet is on**. A helmet on a table, or held in a hand, belongs to nobody, so that worker is still flagged. This part is **our own work**.

2. How it works — six steps in plain language

Think of a supervisor with a checklist who looks at a photo, one step at a time:

#	Step	What happens	File
1	Brighten if dark	If the photo is dark (average brightness below 70 out of 255), we boost the contrast so details are visible. Colours stay true.	`enhance.py`

2	Detect	The AI model draws boxes around every **person**, **helmet** and **vest** it can see, each with a confidence score from 0 to 1.	`detector.py`
3	Match	For each person we decide which helmet and which vest belong to them, using **where** the item is on their body (section 4).	`associate.py`
4	Judge	Apply the rule (helmet and vest required). Each worker gets COMPLIANT, VIOLATION or NEEDS_REVIEW, with a one-sentence reason.	`rules.py`
5	Draw	Green, red or amber boxes and labels on the image.	`annotate.py`
6	Summarise	Counts (workers, compliant, violations), what is missing most, and a downloadable CSV and JSON.	`summary.py`

Only step 2 uses the AI model. Steps 1 and 3–6 are ordinary logic that we wrote and tested ourselves.

3. Words you must know (glossary)

Term	Plain meaning
Bounding box	The rectangle the AI draws around something it found.
YOLO	"You Only Look Once" — a popular, fast family of AI models that find objects in an image in one pass. We use **YOLO11s** (the "s" means small and quick), through the Ultralytics library.
Dataset	A collection of photos where humans already drew the correct boxes and named them. It is the AI's textbook.
Training	Showing the AI the textbook photos over and over so it learns to draw the same boxes on new photos.
Epoch	One full pass through all the training photos. We trained for 60 epochs in the first round, 30 in the second.
Pretrained / transfer learning / fine-tuning	The model we started from had **already** learned to recognise everyday objects from a huge dataset. We did not teach it from zero; we **fine-tuned** it on PPE photos. That is why training took minutes, not days.
Confidence	How sure the AI is about a box, 0 to 1. Our cut-offs: person 0.35, helmet 0.30, vest 0.35.
Precision	Of the things the AI flagged, how many were really there. High precision = few false alarms.
Recall	Of the things that really were there, how many the AI found. High recall = few misses.
mAP@50	One overall score from 0 to 1 that combines precision and recall (a box counts as correct if it overlaps the true box by at least half). Higher is better. 0.87 is a good score on easy data.

Test set	Photos kept hidden from the AI during training. They are the fair exam, because the AI has never seen them.
Head zone / torso zone	Regions of a person's box where a helmet or vest should sit (section 4).
One-to-one matching	Each helmet or vest can belong to at most one person, so one helmet can never "clear" two workers.
NEEDS_REVIEW	Our third outcome: the system is not sure enough to accuse anyone, so it asks a human to check.
CLAHE	A standard contrast-boosting method we apply to dark photos.
Tracking	(Video) Following the same worker from frame to frame so we can talk about "Worker 3" across the whole clip.

4. The core logic we wrote: matching equipment to the right person

Every person box is split into zones, measured from the top of the box:

Zone	Where on the person	What must be there
Head zone	The top 35% of the box (and a little above it, because helmets stick out)	A **helmet**, with its centre inside the zone and at least half of it overlapping the zone
Torso zone	From 20% to 65% down the box	A **vest**, mostly on the person, and big enough (not a tiny patch of colour)

Worked example (say this out loud to be sure you understand)

- **Worker A** has a helmet on their head → it is inside A's head zone → **credited**.
- **Worker B** is holding a helmet at waist height → it is far below B's head zone → **not credited**. B is flagged NO HELMET, and the helmet is shown in grey as "unassigned helmet".
- A helmet lying on a table → inside nobody's head zone → unassigned again.
- One helmet sits between two close workers → it goes to **one** of them only (the one it fits best); the other still needs their own helmet.

> **One-sentence answer for judges:** "We don't ask whether a helmet exists in the picture; we check whether it sits in *that worker's* head zone, and every helmet can only be given to one worker."

Why we don't train a "no helmet" class

A bare head looks very different from person to person and is harder for the AI to spot reliably. It is more robust to detect what *is* there (helmets, vests, people) and **infer** that something is missing when it isn't found on that worker.

5. Three outcomes, not two — and why

If a worker is tiny, cut off by the edge of the photo, hidden in a crowd, or the AI only has a faint guess about their helmet, accusing them would often be wrong. So we say **NEEDS_REVIEW** instead. The six reasons the system uses:

Reason	What triggers it
Very small	The worker is far away (box shorter than 80 px or 8% of the photo)
Cut off	Their head is at the top edge (helmet unverifiable) or their torso is at the bottom edge (vest unverifiable)
Crowded	Their box overlaps another person's box a lot
Unusually wide box	The box is wider than it is tall — possibly several people merged into one

Low person confidence	The AI is unsure this is a person at all
Weak evidence	A helmet or vest was seen on the worker but with confidence between 0.15 and the cut-off. We don't count it as worn, but we also don't accuse the worker

A missing item is only a **VIOLATION** when the worker could be judged fairly. If one item is clearly missing and another is unverifiable, it is still a violation for the missing one.

Poor lighting, distance, occlusion — how we handle them

- **Poor lighting:** dark photos are automatically brightened (CLAHE) before detection.
- **Distance:** we run the AI at a larger image size (960 pixels), and we also scan the photo in four overlapping zoomed-in pieces ("tiles") so small background workers are found. On our own photos this found workers the plain pass missed (for example 7 instead of 5 in one photo). The trade-off: tiling can add a few extra or duplicate boxes, so a person seen only in a tile must be more confident (0.55). Very small workers still become REVIEW.
- **Partial occlusion:** cut-off, crowded or merged workers become REVIEW instead of a false accusation.
- **Multiple workers:** every person gets their own number (left to right) and their own verdict.

6. How we trained the AI (and why it was quick)

	Round 1 (first model)	Round 2 (improved model)
Started from	YOLO11s, pretrained on a large general dataset	Our own round-1 model
Data	Ultralytics *Construction-PPE*: 1,416 photos (1,132 train / 143 validation / 141 test)	Round-1 data plus *CHVG* (1,699 photos): 2,491 train images in total
Classes kept	Three only: person, helmet, vest. Other labels (gloves, boots, "no helmet" etc.) were dropped on purpose.	
Length	60 epochs, **36 minutes**	30 epochs, **40 minutes**
Hardware	Laptop GPU: NVIDIA RTX 5060, image size 960 px	
Extras	Stronger brightness variation during training, so the model copes with poor lighting	

> **Why was training quick?** "We used transfer learning. The model already knew how to find people and objects, so we only had to teach it three PPE classes. That takes minutes to an hour on a laptop GPU, not days."

Where the data came from (credit it!)

- **Construction-PPE** dataset — Ultralytics, licence AGPL-3.0.
- **CHVG** dataset — Md. Ferdous and Sk. Md. Masudul Ahsan, 2022, *PeerJ Computer Science 8:e999*, licence CC BY 4.0.
- **Ultralytics YOLO**, OpenCV, Streamlit, pandas — open-source libraries.

7. Our results — what we can honestly say

We scored the model on photos it never trained on. We used **two different test sets** on purpose.

Test set	Round 1 model	Round 2 model
Original test photos (141) — same source as the training photos	mAP@50 **0.866** (precision 0.890, recall 0.862)	mAP@50 **0.829** (precision 0.895, recall 0.841)
New photos from other sites, CHVG (173) — the **fairer** test	mAP@50 **0.758**	mAP@50 **0.882** (precision 0.896, recall 0.892) — see the note below

Per class (round 1, original test photos): helmet 0.92, vest 0.87, person 0.81. **Person detection is our weakest class.**

Read the numbers carefully — this is the honest version.
• The round 1 model scored 0.866 on photos that look like its training photos, but only **0.758** on photos from other sites. That gap is why we added a second dataset.
• The round 2 model scored **0.882** on the CHVG test photos, but it trained on the *training half* of that same collection, so that is not a like-for-like comparison with round 1. On the original test photos it went **down a little** (0.866 to 0.829).
• What we *can* say: round 2 saw far more varied photos, and on our own real site photos it found more workers and made fewer mistakes (for example it found a worker with a yellow helmet that round 1 missed, and correctly flagged a worker with no vest that round 1 called compliant).
• If a judge asks "what's your accuracy?", give the numbers *with what they were measured on*. That earns more trust than one big number.

8. What works, what doesn't (our known limits)

Works well	Still weak (say this before the judges find it)
• Clear photos of workers with or without helmet and vest • Several workers in one photo, each with their own verdict • A helmet held in hand or on a table is **not** credited (unit-tested) • A cap is not mistaken for a helmet (seen on a real photo) • Crowded or unclear workers become REVIEW, not false accusations • Dark photos are brightened automatically	• Small or background workers are sometimes missed; zoomed-in tiling finds more but can add a duplicate box on one person • Seated or bending workers may get REVIEW instead of VIOLATION (their box is wide) • Orange life jackets or hi-vis shirts can be mistaken for a safety vest • Two people close together can be merged into one box • We have not yet measured the "hard cases" (helmet in hand, helmet on table, hi-vis non-vest, dark, distant) on a full test set of our own photos

9. Who explains what — five roles

Split these among yourselves. Everyone should read all of them once.

Role A — The problem and the big picture

Likely question	Short answer
What problem are you solving?	Supervisors can't watch every worker. We automatically check each worker for helmet and vest and flag violations.
What is the input and the output?	Input: a photo (or short clip). Output: the same image with a green, red or amber verdict per worker, plus a table, a violation summary and CSV/JSON files.
What makes your solution different?	We match equipment to the *specific* worker wearing it, so a helmet elsewhere in the image can't clear everyone.
Could this be used in a real factory?	It's an MVP. It would need more training on that site's own photos and camera angles, and a human reviewing the amber cases.

Role B — Data and training

Likely question	Short answer
Which data did you use?	Two public datasets: Ultralytics Construction-PPE and CHVG, both credited. About 3,100 photos in total.
Did you train from scratch?	No. We fine-tuned a pretrained YOLO11s. That is why it took under an hour per round.
Why only three classes?	Person, helmet and vest are all we need. We infer "missing" from absence instead of training a "no helmet" class.
What hardware did you use?	A laptop with an NVIDIA RTX 5060 GPU.
How do you know it isn't just memorising?	We test on photos it never saw, including a second dataset from different sites, where the score is lower and more realistic.

Role C — Matching logic (the core)

Likely question	Short answer
How do you know a helmet belongs to this worker?	Each worker has a head zone (top 35% of their box). A helmet counts only if its centre is in that zone and most of it overlaps the zone.
What stops one helmet clearing everyone?	One-to-one matching: each helmet and vest is assigned to at most one worker.
What about a helmet in hand or on a table?	It is outside every head zone, so nobody gets credit. The worker is flagged, and the helmet is shown as "unassigned".
How do you stop a small orange patch counting as a vest?	The vest must be mostly on the person, centred on the torso, and at least 45% of the person's width and 20% of their height.

Role D — Rules, app and outputs

Likely question	Short answer
What are the compliance rules?	Helmet and vest required by default. It's a setting, so a site can require only a helmet.
What does NEEDS_REVIEW mean?	The worker is too small, cut off, crowded, or the evidence is weak. We don't accuse; we ask a human to check.
How do you handle poor lighting and distance?	Dark photos are brightened automatically; we run at a larger image size; tiny workers go to REVIEW.
What does the app show?	The annotated image, counts, a per-worker table, the reason for each decision, and download buttons for PNG, CSV and JSON. There is also a video page and a batch analyser for a whole folder.
What about video?	Each worker is tracked, and a violation is reported only after it persists for 5 of the last 8 usable frames, so one-frame flicker doesn't cause false alarms.

Role E — Testing, results and honesty

Likely question	Short answer
How did you test it?	69 automatic tests check the matching and rules with hand-made boxes, plus an evaluation script that scores our own photos against an answer key.
What is your accuracy?	Give the numbers from section 7 with what each was measured on, and explain why round 2's 0.882 is not a like-for-like comparison.
Where does it fail?	Small or background workers can be missed; seated or bending workers may come out as REVIEW; life jackets can look like vests. (Read section 8.)
What would you do with more time?	More training photos, especially hard cases (helmet in hand, non-vest hi-vis), tune thresholds on our own test photos, and add more camera angles.
Did you write the code yourselves?	The matching, rules, summary, app, tests and evaluation are our own work. We used YOLO and public datasets for detection, and Claude as a coding assistant, which we reviewed and understand.

10. Rules for answering judges

Do	Don't

• Answer in one or two sentences, then stop • Use the analogy: "a supervisor with a checklist" • Say "we fine-tuned a pretrained model" • Give accuracy **with what it was measured on** • Admit limits first, then say how you would fix them • Say "I don't know, but [teammate] does" when needed	• Don't say "we trained it from scratch" • Don't say "we collected the data" • Don't say "100% accurate" or quote one big number alone • Don't claim it handles every hard case • Don't guess technical details — say what you do know • Don't hide that Claude helped; it is credited in the README

11. The 60-second pitch (everyone learn this)

> "Supervisors can't watch every worker, so unsafe workers get missed. Our system takes a photo, finds every worker, and checks **that worker** for a helmet and a vest. A helmet on a table or in someone's hand doesn't count, because we match each item to the person wearing it. Each worker gets a green, red or amber verdict — amber when we're not sure, so we never accuse someone we can't see properly — and we produce a violation summary you can download. We fine-tuned a pretrained YOLO model on two public datasets, and wrote the matching, rules, summary and app ourselves. Our accuracy is [fill in: both numbers], and we know where it still fails: small background workers and look-alike clothing."

12. Quick facts to memorise

Model	YOLO11s (Ultralytics), fine-tuned
Classes	person, helmet, vest
Datasets	Construction-PPE (1,416 photos) + CHVG (1,699 photos)
Training	Round 1: 60 epochs, 36 min. Round 2: 30 epochs on the merged data
Image size	960 pixels, plus 2x2 zoomed tiles for small workers
Head zone / torso zone	Top 35% of the person box / 20%–65% down
Confidence cut-offs	Person 0.35, helmet 0.30, vest 0.35; weak evidence from 0.15
Verdicts	COMPLIANT (green), VIOLATION (red), NEEDS_REVIEW (amber)
Video rule	Confirm a violation after 5 of the last 8 usable frames
Tests	69 automatic tests, all passing
How to run the app	`streamlit run app.py` then open `http://localhost:8501`